Maximizing Pollination Services In Urban Orchards

Preliminary Report for SEED Saint Louis

1 December 2025

Introduction

With the world's population growing, we need to produce more food without harming the environment. However, growing food in densely populated areas poses challenges, especially ensuring sufficient insect pollinators for fruiting success. For this project, we were interested in understanding (1) how and why pollination varies across community orchards in St. Louis and (2) whether or not intentionally increasing pollinators or pollinator habitat promotes fruit production. To do this, we looked at pollination from the scale of the individual flower to the whole orchard, video recorded pollinator behaviors, took weekly flower and pollinator collections, and tracked fruit production over four years.

This research was only possible because it was a community effort! Our team included professionals and students from four local universities, the Missouri Botanical Garden, the Saint Louis Zoo, and SEED St. Louis. We are incredibly thankful to the folks at 18 different community and commercial orchards (shown in the image below) for working with us. Ultimately, our goal is to inform best practices for maximizing urban orchard economic value and environmental benefits, supporting local insect diversity and minimizing cities' ecological footprint.With your help, we are one step closer to achieving that goal. This report provides a summary of our findings to date.



Link to google map

Project Summary

Project goals: To understand (1) how and why pollination varies across community orchards in St. Louis and (2) whether or not intentionally increasing pollinators or pollinator habitat promotes fruit production. To address our two research goals, we collected a lot of different types of data. Here we summarize some those data in four main sections:

1. Who is pollinating your orchard trees?
2. What are your pollinators doing?
3. Are your orchard trees receiving enough pollen?
4. What impact do the above have on fruit production?

Each section required different methods, which are outlined below to clarify how we compiled our results.

1. **Who is pollinating your orchard trees?.....pg 5**
   Every few days, we visited each orchard and collected insects on all orchard trees that were in bloom using insect nets. We then identified those insects to determine who and how many individuals were visiting.

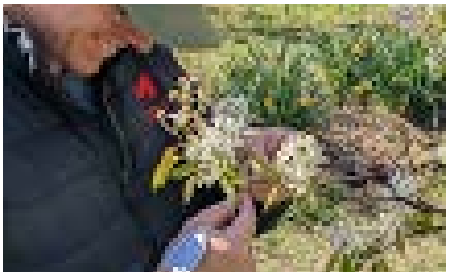
2. **What are your pollinators doing?.....pg 6**
   We used camcorders to record insect behavior on small clumps of flowers about every 10 days (weather depending) throughout the blooming season. Each recording lasted between an hour and two hours. We then watched the videos to record what, exactly, they were doing on the flowers. How were they collecting nectar and/or pollen? Do some species act differently than others?

3. **Are your orchard trees receiving enough pollen?.....pg 8**
   On each tree, we tagged several pairs of flowers. One flower is left untouched, which tells us how much pollen plants are receiving from pollinators. The other flower is left open to pollinators but we also add pollen by hand to provide as much pollen as possible to the flower. We then compare how many pollen grains grow tubes to the ovary, which could later turn into fruit.

4. **What impact do the above have on fruit production?.....pg 10**
   For this question, we captured bees as they were pollinating, and anesthetized them by cooling them down on ice. Then, we used the sleepy bees to pollinate the flowers, and tested which pollinator visits resulted in the largest fruits.

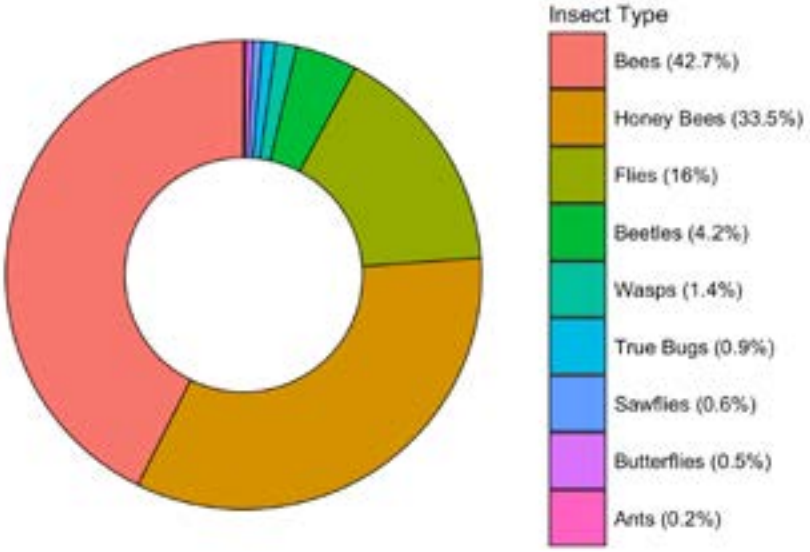

George Todd netting insects on a tree.


Camcorder recording insect activity.


Javier De Soto tagging flowers.


An aensthetized bee pollinating a flower.

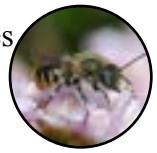
Who is pollinating your orchard trees?



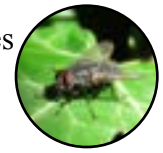
Bees were the most abundant foragers on orchard trees at these 18 urban orchards and farms. Honey bees (*Apis mellifera*) alone accounted for 33.5% of the insects collected. All other bee species, such as mining bees (*Andrena* spp.) and mason bees (*Osmia* spp.), made up 42.7%. Importantly, the next most abundant pollinators were flies, which made up 16% of the observations. Flies help fill the gap in pollination when bees are less active during cool, cloudy, or early spring days.

Beetles (4.2%), wasps (1.4%), true bugs (0.9%), sawflies (0.6%), butterflies (0.5%), and ants (0.2%) were also pollinating orchard trees, but in smaller numbers. Despite their smaller numbers, these other insect groups still contribute to the diversity of flower visitors. Even with low numbers, rarer species can be important pollinators, because they provide additional pollen movement.

Other studies have shown that having higher pollinator diversity increases fruit yield. With greater pollinator diversity, you can get more consistent pollination throughout the season. Even if one group of insects isn't as active because of weather or changes in their habitat, others can compensate, potentially make fruit yeild more reliable. We have yet to test this with our data, but that is one of the next analyses that we have planned.

Bees 

Honey Bees 

Flies 

Beetles 

Wasps 

True Bugs 

Sawflies 

Butterflies 

Ants 

CREDIT: https://www.flickr.com/photos/63075200@N07/albums/
everything is from this person flickr | Steven Falk

What are your pollinators doing?

Most of the insects that visited the orchards are active during the day, but their peak activity depends on the group and the weather. Bees, wasps, butterflies and beetles are most active on warm, sunny days because they need to be warm up in order to fly. When it is cool and overcast, they are less active. Flies, on the other hand, are often active earlier in the morning and under cooler, cloudier, or even slightly rainy conditions, making them especially important pollinators in the spring.

Different kinds of fruit trees flower at different times of the year. Similarly, different pollinators are also active at different times of year. The figures on the next page show the day of the year when each tree type (page 7 top) and pollinator group (page 7 bottom) is most active. Early flowers attract early insects like flies and honey bees, whereas mid-season blossoms match the highest variety of active pollinators. Late-blooming trees are more likely to be visited by bumble bee queens, beetles, true bugs and sawflies. These results demonstrate the importance of having a diverse pollinator community. Each group of pollinators peaks at a different time, ensuring that plants continue to be pollianted throughout the flowering season.

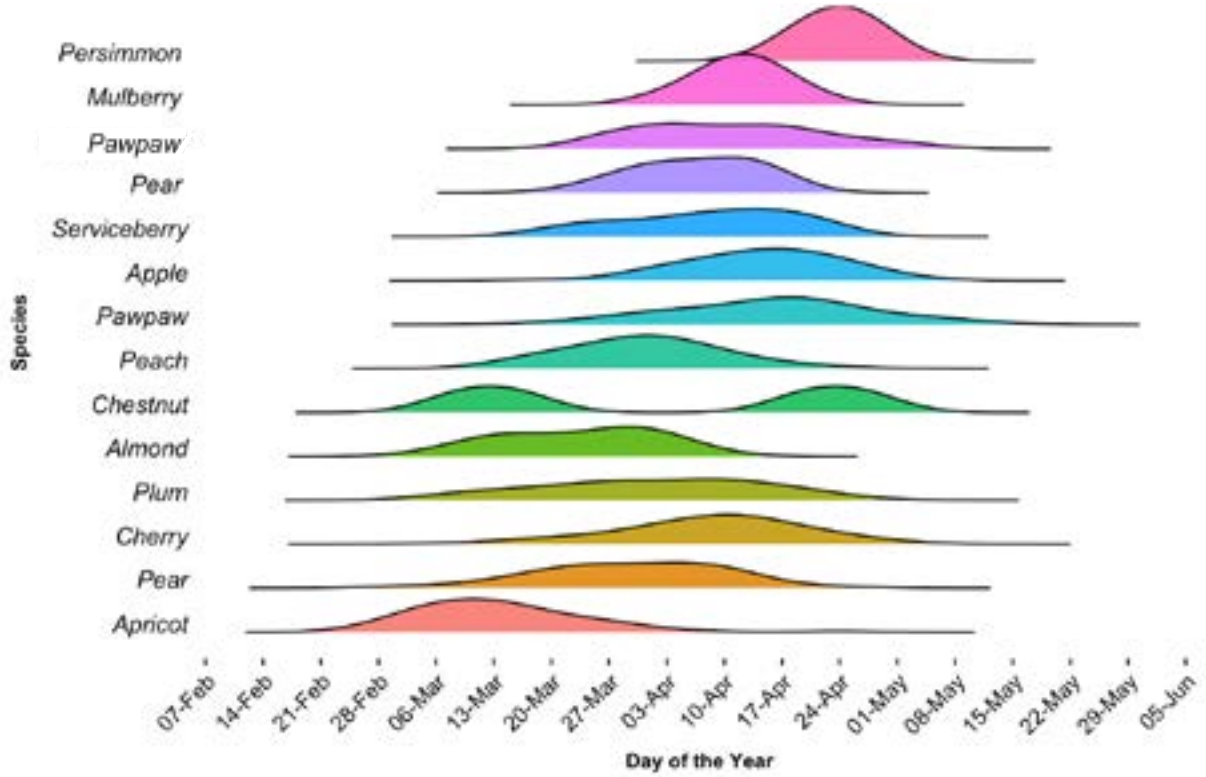
Pollinator diversity is also important, because each group behaves a little bit differently. Using our video recordings, we observed bees and flies and recorded how they gathered pollen and nectar from the flowers. We also noted whether or not they touched the reproductive parts of the flower. It turns out that flies, while abundant spend much more time standing on the floewr or grooming themselves compared to bees. Though these behaviors may reduce the amount of time the flies are touching the reproductive parts of the flower, they can still transmit pollen effectively due to their hairy bodies and how long they stay on a given flower.

Different types of bees behave differently as well. Some solitary bee groups (such as the Andrenids) are more likely to stick their entire head directly into the middle of the flower to drink nectar at the base. This ensures maximum contact with the reproductive parts. On the other hand, honey bees tend to be fast in their visits and will often try to drink nectar by inserting their heads into the flower from a side angle which means that they avoid touching the reproductive parts.

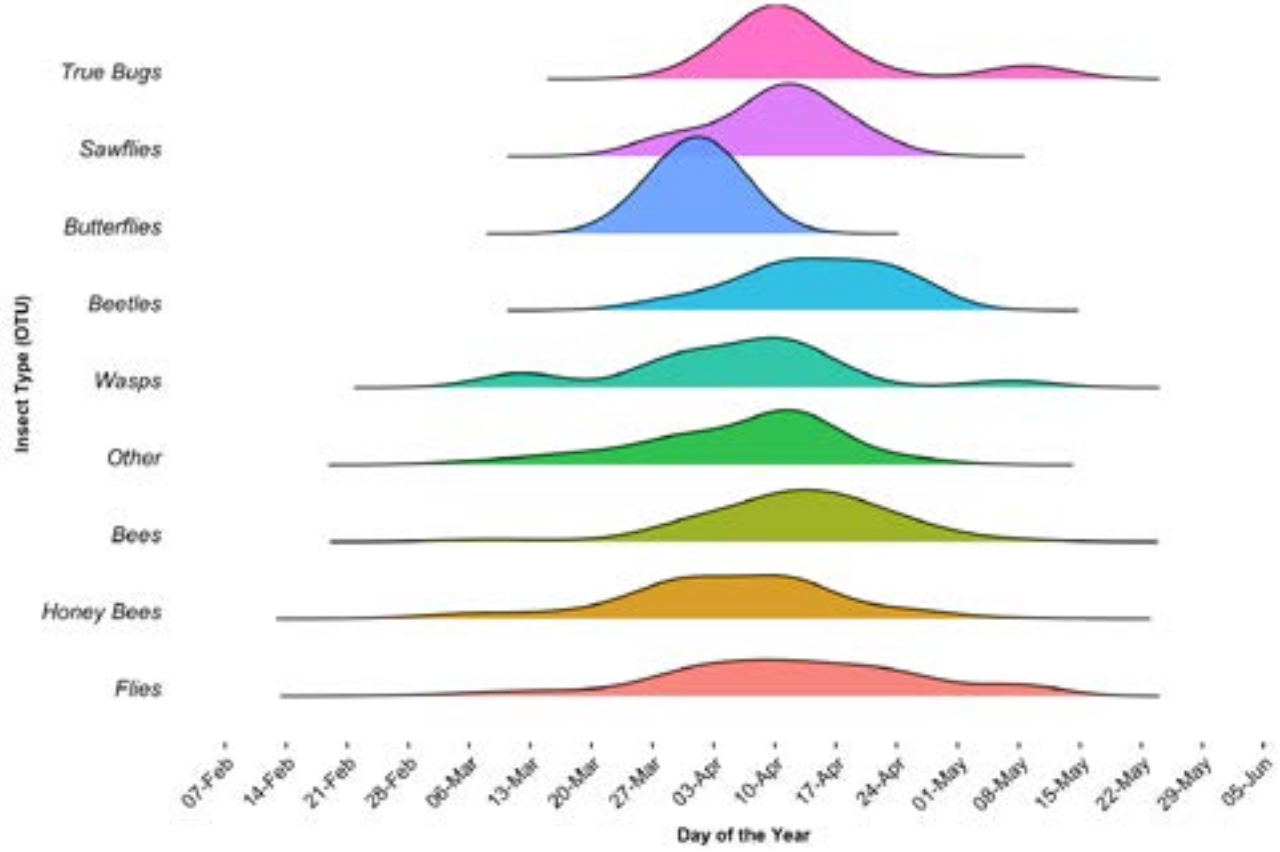
Finally, bee behavior changes as you go from the suburbs into the city. In the city, honey bees are more likely to try a side angle approach to drinking nectar from flowers, which means they are less likely to touch the reproductive parts of the flower. Andrenids are more likely to remain stationary on flowers that are of moderate urbanness, so they are spending less time pollinating the flower. Our future analyses will tease apart what other factors (such as the quantity and type of trees at the sites) are influencing which behaviors we see on the flowers. Because both visit duration and specific behaviors can vary widely, having healthy pollinator diversity can mitigate any negative effects of a single pollinator group.

Left image: QR code for videos of bees displaying some nectar and pollen foraging behaviors. We recommend watching with the video on mute, becasue the slower speed messes up the audio. You can also find it here: https://www.youtube.com/watch?v=EItaQwSUE-c

(a) **Flowering Schedules by Tree Type**



(b) **Insect Activity by Insect Type**



These figures show (a) the flowering schedules for 14 orchard trees and shrubs and (b) the activity schedules for 9 types of insects. Peaks indicate the dates on which the most individuals trees were flowering or insects were foraging. The flowering schedules of each species aligned best with different types of pollinators, suggesting that orchard plants rely on different pollinators throughout the flowering season.

Are your orchard trees receiving enough pollen?

Flowers often do not receive enough pollen to produce as many fruits as they are capable of. This is known as pollen limitation. Fruit and seed set can be constraied by an inadequate quantity or quality of pollen. It can happen when pollinators don't deliver enough pollen, when pollen comes from the wrong species or plant, or when environmental stress disrupts pollinator activity. In short, the plant has viable ovules waiting, but not enough of the right pollen reaches the stigma to complete fertilization.

From the plant's perspective, pollination seems simple: attract a pollinator, get pollen to stick to it, and have that pollen carried to another flower. Once there, the pollen grain germinates and grows a pollen tube down into the ovary to deliver sperm to the egg. When everything works, we see healthy fruit set. When it doesn't, we see pollen limitation. In orchard systems, where most fruit trees rely on wild bees and other insects for pollination, pollen limitation provides a direct measure of how well pollination services are functioning.

Urban orchards depend on a diverse community of bees and other insects to move pollen among many different tree species. If diversity or abundance drops—due to habitat loss, fragmented landscapes, or reduced nesting resources—pollination efficiency may decline. Detecting pollen limitation helps us see exactly where those breaks in the system occur.

We test for pollen limitation using pollination supplementation experiments.

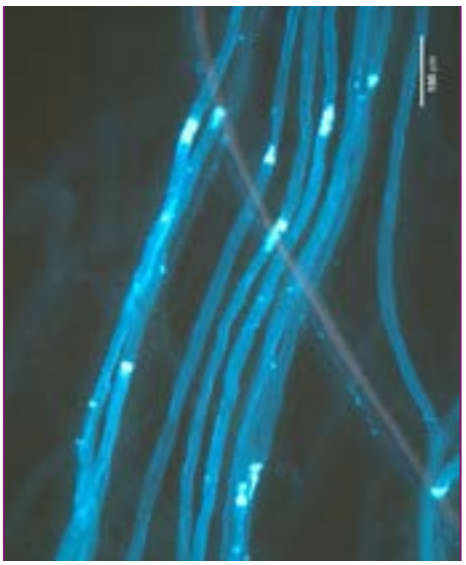
1. We leave some flowers to open pollination under natural field conditions.
2. We hand-apply pollen to other flowers (known as supplementation), giving them all the compatible pollen they could want.

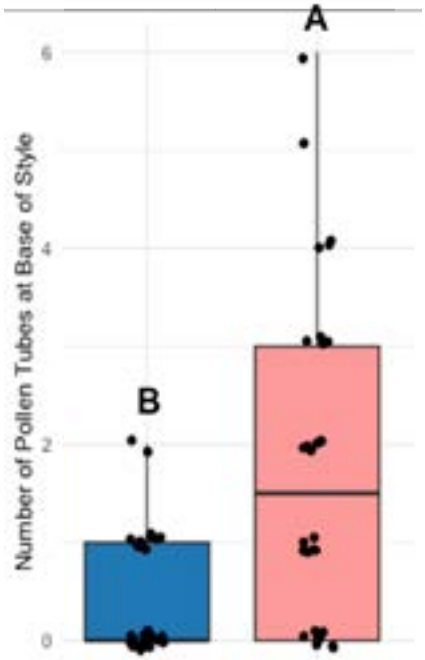
After pollination, we collect pistils, stain them, and count pollen tubes growing down the style to the ovary using a florescent microscope (below left). Comparing pollen tube growth among these treatments tells us whether natural pollination is sufficient—or whether the plants are pollen-limited.

Open pollination



Supplementation





The orchard trees were indeed pollen limited (above), suggesting that orchard trees are not receiving enough pollen from wild pollinators. Efforts to increase pollinator abundance or diversity may reduce pollen limitation and increase fruit production.

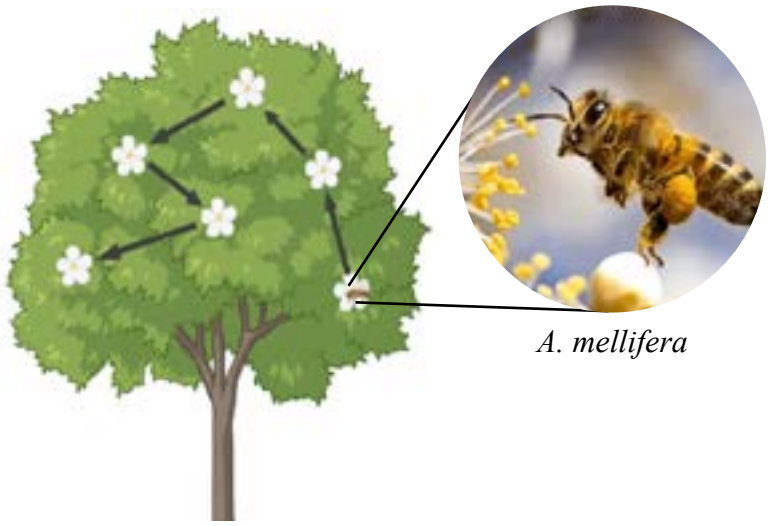
What impact do the above have on fruit production?

We looked at apple pollination in urban orchards in St. Louis to see what might be stopping fruit production. We wanted to know if some bee species were better pollinators than others. To do this, we put bees to "sleep" by cooling them down. Then, we brushed their bodies up against a receptive stigma of an apple flower. At the end of the growing season, we collected the fruits to see which bee species was the most successfull pollinator.

We found that honeybees visit apple flowers often, but they usually stay on the same tree instead of moving between trees (left). This means that pollen from the same tree was morely to pollinate the flowers.

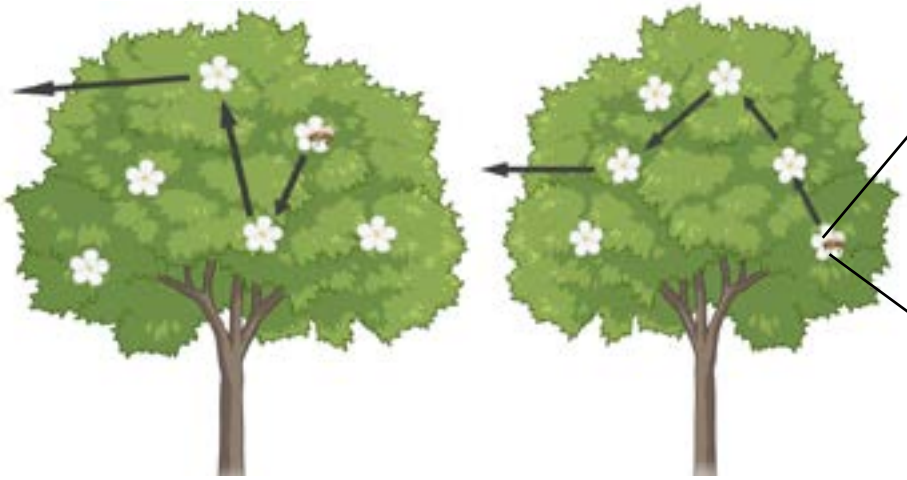
Native bees, like mining bees (*Andrena ssp.*) and mason bees (*Osmia ssp.*), were better at moving pollen between trees (below), so they helped to make more apples.

When plants receive self-pollen, they are often smaller or produce fewer seeds. We see that here, as well. Few apples grew to maturity when pollinated by honey bees than when pollinated by mason bees.



A. mellifera

Tree A - Honey bees



O. cornifrons



Andrena sp.

The foraging behavior of mining bees (*Andrena ssp.*) and mason bees (*Osmia cornifrons*) in apple orchard. The arrows in the images depict the movement of these bees on the trees. Honey bees move more frequently between flowers on the same tree. Mining and mason bees move more pollen between trees. This cross-tree movement promotes cross-pollination, which enhances the likelihood of successful apple fruit development. Mason bees, in particular, are active early in the year and and often carry substantial amounts of pollen, making them particularly valuable contributors to apple fruit set alongside mining bees.

What is next?

Our data collection is now complete, and we have documented some important patterns regarding pollination in St. Louis' orchards. However, there are still several analyses to run before we have a complete understanding of what these data mean for pollinators and fruit production in St. Louis community orchards. We are diligently working on identifying all of our insect specimens to species-level, so we can see if different species within a pollinator group respond the same way to the urban environment. We are analyzing our videos from the plant perspective to determine if plants are receiving poorer pollination when pollinators behave certain ways. We are testing results from an experiment that we ran to increase pollinator abundances in a subset of orchards. Ultimately, our goal is analyze all of these data together to determine which factors resulted in the highest fruit yield. We will continue to share our new findings and what they mean for your orchards as they emerge.

Acknowledgements

First, we want to thank the managers and volunteers of the community orchards and urban farms! They've been incredibly supportive throughout the implementation of the grant. We simply could not do this work with their help. Thank you to everyone involved with the following community gardens and farms: 13th Street Community Garden, Blessings and Glory, Botanical Heights Community Garden, Carondelet Food Forest, Circle of Light Associates Victory Garden, EarthDance Organic Teaching Farm, Emmanuel Church Orchard, Florissant Community Garden, Herman's Farm, House of Living Stone Community Garden, Kellogg Park Community Garden and Orchard, McKinley Meadows Community Garden, Old Ferguson West Community Garden, Our Lady of the Holy Cross Catholic Church Orchard, Rustic Roots Sanctuary, St. Louis University Teaching Garden and Orchard, The Orchard on Virginia, and Thies Farm and Greenhouses, Inc.

Special thanks go to Dean Gunderson and SEED St. Louis who have been involved in the grant since the very beginning. We also want the thank the amanzing undergraduate and graduate students who contributed to this work. This is an ambitious project that looks at pollination from many different ecological angles and spatial scales; student work has been critical to the success of the project.

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